Lesson 3.10 — Unit 3 Review and Assessment

Big idea: Unit 3 is everything about polynomials: classify them, predict their shape from end behavior + zeros, divide and factor them, and use the Remainder/Factor theorems to fill in unknowns.

Quick review

Identify: degree and leading coefficient → end behavior. Even/odd degree, ± leading.
Zeros: use factoring or RRT to find rational zeros, then divide. Multiplicities decide cross vs. bounce.
Division: long for any divisor, synthetic for linear (x – c).
Theorems: Remainder (P(c) = remainder), Factor (P(c) = 0 ■ (x – c) is a factor), FTA (degree n → n complex zeros).
Conjugate pairs: non-real zeros of real polynomials come in pairs.

Worked example

Worked review. $P(x) = x^3 - 6x^2 + 11x - 6$. Test x = 1: 1 – 6 + 11 – 6 = 0 ✓. Divide: $x^2 - 5x + 6 = (x - 2)(x - 3)$. Zeros: 1, 2, 3.

Warm-ups

1. State end behavior of $P(x) = -2x^5 + 3x^2$.
2. Find P(–1) for $P(x) = x^4 + 2x - 3$.
3. Is (x + 4) a factor of $P(x) = x^2 - 16$?

Practice

1. Factor $x^3 - 2x^2 - x + 2$ completely.
2. Use synthetic division to divide $x^3 + 4x^2 - x - 4$ by (x + 1).
3. Find all real zeros of $x^3 - x^2 - 4x + 4$ (test small candidates).

Challenge

1. Write a degree-4 polynomial with real coefficients whose zeros are 1, –2, and 3 + i. Then expand to standard form.
2. $P(x) = ax^3 + bx^2 - x + 2$. If P(1) = 4 and (x – 2) is a factor, find a and b.

Answer key — Lesson 3.10

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Degree 5 (odd), leading – → up/down (left up, right down).
2. $1 - 2 - 3 = -4$.
3. $P(-4) = 16 - 16 = 0 \Rightarrow$ yes.

Practice

1. Group: $x^2(x - 2) - (x - 2) = (x - 2)(x^2 - 1) = (x - 2)(x - 1)(x + 1)$.
2. –1 **|** 1 4 –1 –4 → row –1, –3, 4; sums 1, 3, –4, 0. Quotient: $x^2 + 3x - 4 = (x + 4)(x - 1)$. So P factors as $(x + 1)(x + 4)(x - 1)$.
3. Test $x = 1$: $1 - 1 - 4 + 4 = 0$ ✓. Divide → $x^2 - 4 = (x - 2)(x + 2)$. Zeros: 1, 2, –2.

Challenge

1. Conjugate: $3 - i$. $P(x) = (x - 1)(x + 2)(x - (3+i))(x - (3-i)) = (x^2 + x - 2)(x^2 - 6x + 10) = x^4 - 5x^3 + 2x^2 + 22x - 20$.
2. $P(1) = a + b - 1 + 2 = a + b + 1 = 4 \Rightarrow a + b = 3$. $P(2) = 8a + 4b - 2 + 2 = 8a + 4b = 0 \Rightarrow 2a + b = 0$. Subtract: $a = -3$, $b = 6$.